

PostgreSQL Portland Performance Practice Project

Database Test 2 (DBT-2)
Characterizing Filesystems

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Review

From last time:

- ▶ DBT-2 live demo.

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```

This is:

- ▶ A collection of reason why you should test your own system.
- ▶ A guide for how to understand the i/o behavior of your own system.
- ▶ A model of simple i/o behavior based on PostgreSQL

This is not:

- ▶ A comprehensive Linux filesystem evaluation.
- ▶ A filesystem tuning guide (mkfs and mount options).
- ▶ Representative of your system (unless you have the exact same hardware.)
- ▶ A test of reliability or durability.

Contents

- ▶ Criteria
- ▶ Preconceptions
- ▶ fio Parameters
- ▶ Testing
- ▶ Test Results

Some hardware details

- ▶ HP DL380 G5
- ▶ 2 x Quad Core Intel(R) Xeon(R) E5405
- ▶ HP 32GB Fully Buffered DIMM PC2-5300 8x4GB DR LP Memory
- ▶ HP Smart Array P800 Controller
- ▶ MSA70 with 25 x HP 72GB Hot Plug 2.5 SAS 15,000 rpm Hard Drives

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```

Criteria

PostgreSQL dictates the following (even though some of them are actually configurable:

- ▶ 8KB block size
- ▶ No use of `posix_fadvise`
- ▶ No use of direct i/o
- ▶ No use of async i/o
- ▶ Uses processes, not threads.

Based on the hardware we determine these criteria:

- ▶ 64GB data set - To minimize any caching in the 32GB of system memory.
- ▶ 8 processes performing i/o - One fio process per core.

Linux Filesystems Tested

- ▶ ext2
- ▶ ext3
- ▶ ext4
- ▶ jfs
- ▶ reiserfs
- ▶ xfs

```
a8888b. . o 0 ( What is your favorite filesystem? )
d888888b.
8P"YP"Y88
8|o||o|88
8'      .88
8'...' Y8.
d/      '8b.
.dP      . Y8b.
d8:'      " ' :88b.
d8"      'Y88b
:8P      ' :888
8a.      : _a88P
._/"Yaa_ : .| 88P|
\      YP" ' | 8P '
/      \._____d|      .'
'---..__)888888P'...'
```

Hardware RAID Configurations

- ▶ RAID 0
- ▶ RAID 1
- ▶ RAID 1+0
- ▶ RAID 5
- ▶ RAID 6

```
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```

Preconceptions

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   |__|      |__| -"

```

. o 0 (Everyone gets to participate.)

Flexible I/O by Jens Axboe.

- ▶ Sequential Read
- ▶ Sequential Write
- ▶ Random Read
- ▶ Random Write
- ▶ Random Read/Write

¹<http://brick.kernel.dk/snaps/>

Sequential Read

```
[seq-read]
rw=read
size=8g
directory=/test
fadvise_hint=0
blocksize=8k
direct=0
numjobs=8
nrfiles=1
runtime=1h
exec_prerun=echo 3 > /proc/sys/vm/drop_caches
```

Sequential Write

```
[seq-write]  
rw=write  
size=8g  
directory=/test  
fadvise_hint=0  
blocksize=8k  
direct=0  
numjobs=8  
nrfiles=1  
runtime=1h  
exec_prerun=echo 3 > /proc/sys/vm/drop_caches
```

Random Read

```
[random-read]
rw=randread
size=8g
directory=/test
fadvise_hint=0
blocksize=8k
direct=0
numjobs=8
nrfiles=1
runtime=1h
exec_prerun=echo 3 > /proc/sys/vm/drop_caches
```

Random Write

```
[random-write]  
rw=randwrite  
size=8g  
directory=/test  
fadvise_hint=0  
blocksize=8k  
direct=0  
numjobs=8  
nrfiles=1  
runtime=1h  
exec_prerun=echo 3 > /proc/sys/vm/drop_caches
```

Random Read/Write

```
[read-write]
rw=rw
rwmixread=50
size=8g
directory=/test
fadvise_hint=0
blocksize=8k
direct=0
numjobs=8
nrfiles=1
runtime=1h
exec_prerun=echo 3 > /proc/sys/vm/drop_caches
```

Versions Tested

Linux 2.6.25-gentoo-r6

fio v1.22

and

Linux 2.6.28-gentoo

fio v1.23

```

      --      --
    /  \  ~~~ /  \   . o O ( Results! )
  ,----- (      oo      )
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   |__|    |__| -"

```

There is more information than what is in this presentation, raw data is at: <http://207.173.203.223/~markwkm/community6/fio/2.6.28/>
For example other things of interest may be:

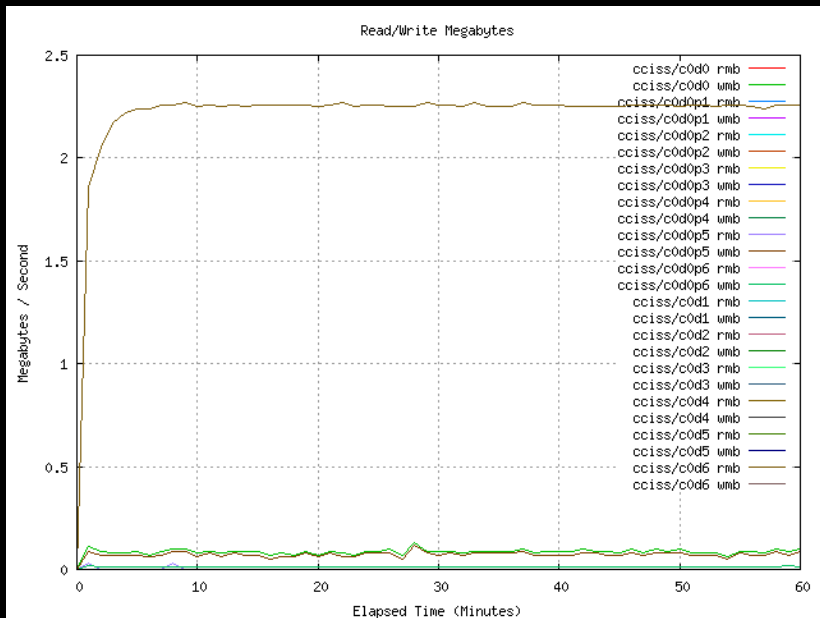
- ▶ Number of i/o operations.
- ▶ Processor utilization.

A few words about statistics...

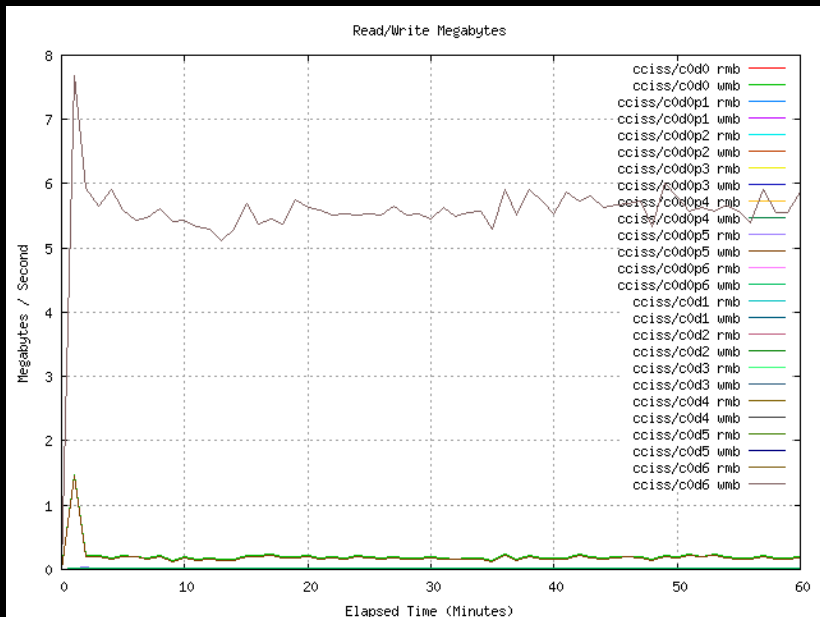
Not enough data to calculate good statistics.

Examine the following charts for a feel of how reliable the results may be.

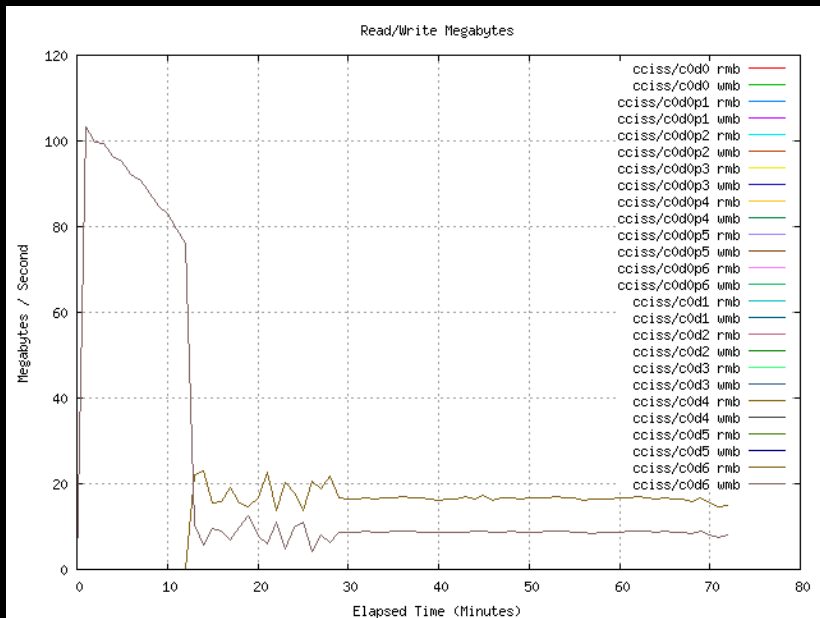
Random Read with Linux 2.6.28-gentoo ext2 on 1 Disk



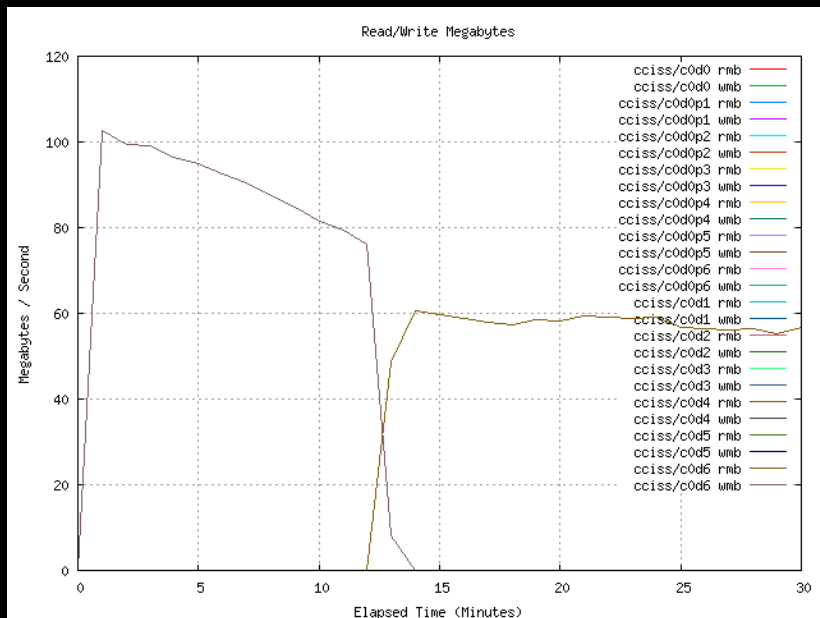
Random Write with Linux 2.6.28-gentoo ext2 on 1 Disk



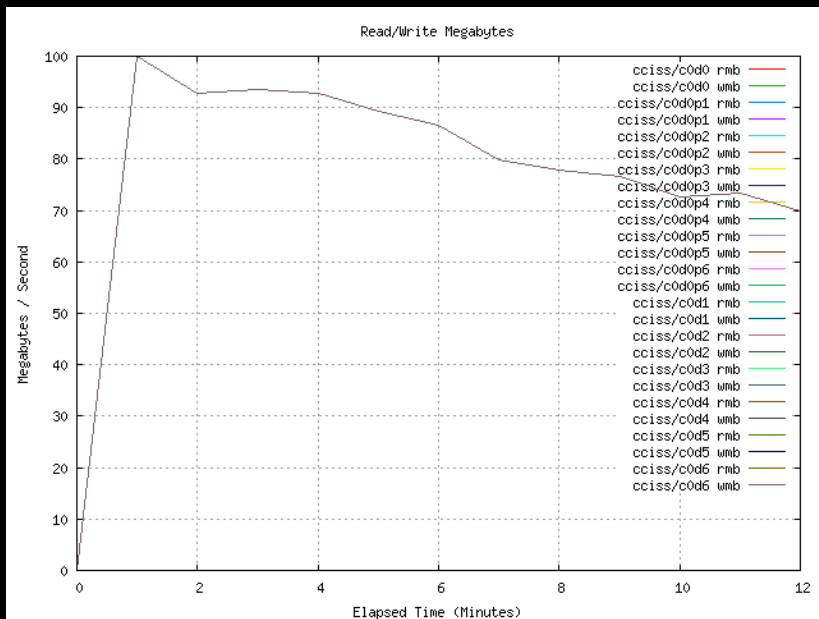
Read/Write Mix with Linux 2.6.28-gentoo ext2 on 1 Disk



Sequential Read with Linux 2.6.28-gentoo ext2 on 1 Disk

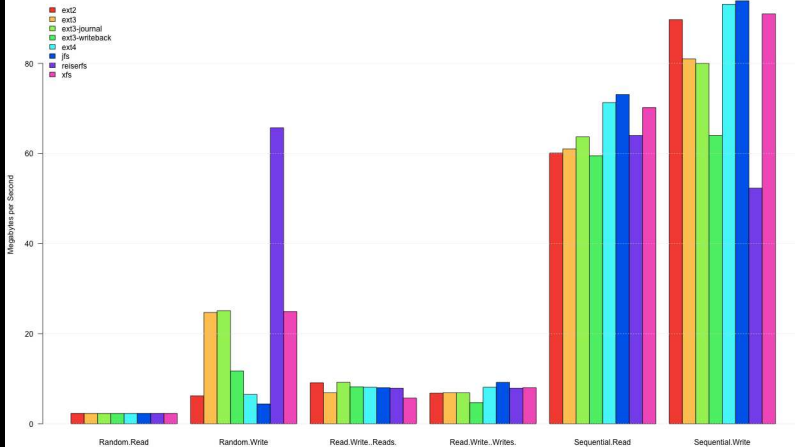


Sequential Write with Linux 2.6.28-gentoo ext2 on 1 Disk



Results for all the filesystems when using only a single disk...
(If you see missing data, it means the test didn't complete.)

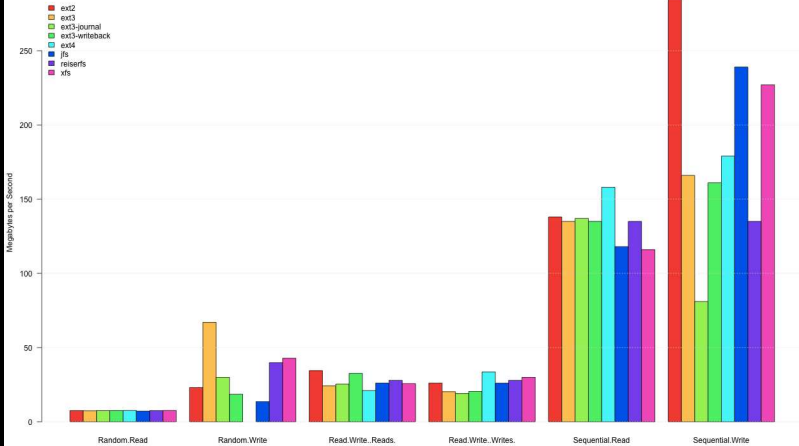
1 Disk RAID 0



2.6.28-gentoo

Results for all the filesystems when striping across four disks...

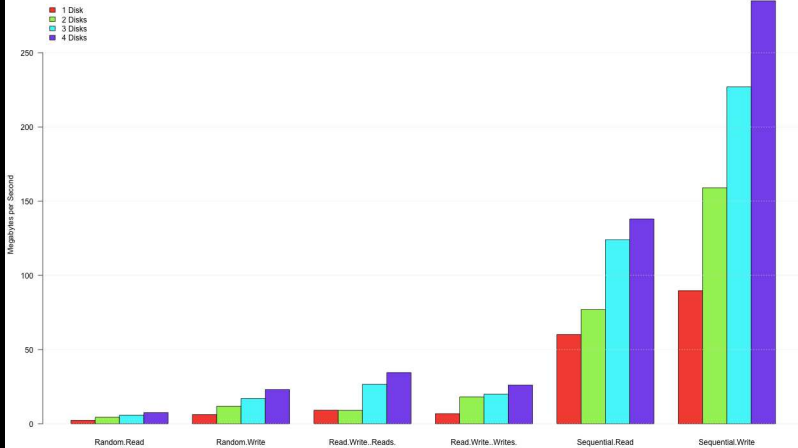
4 Disk RAID 0



2.6.28-gentoo

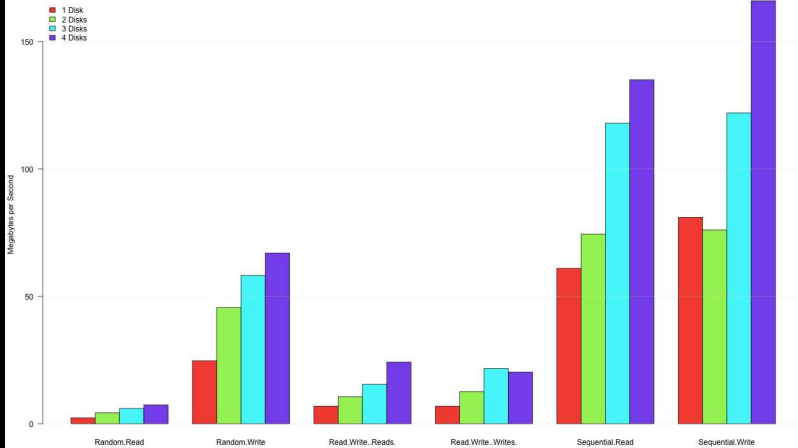
Let's look at that step by step for RAID 0.

ext2 RAID 0



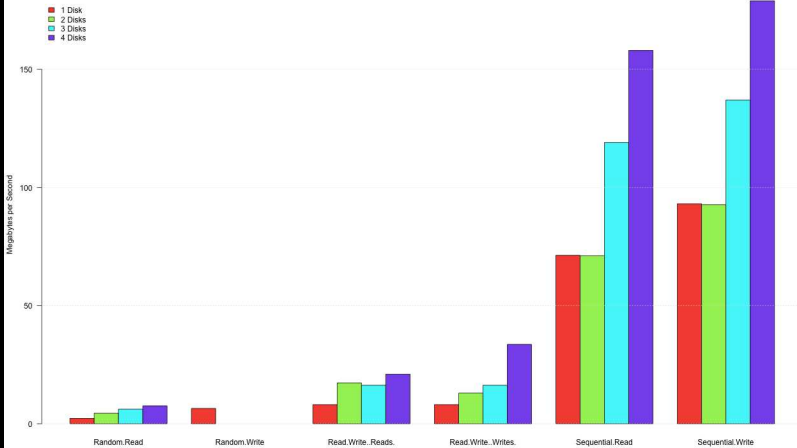
2.6.28-gentoo

ext3 RAID 0



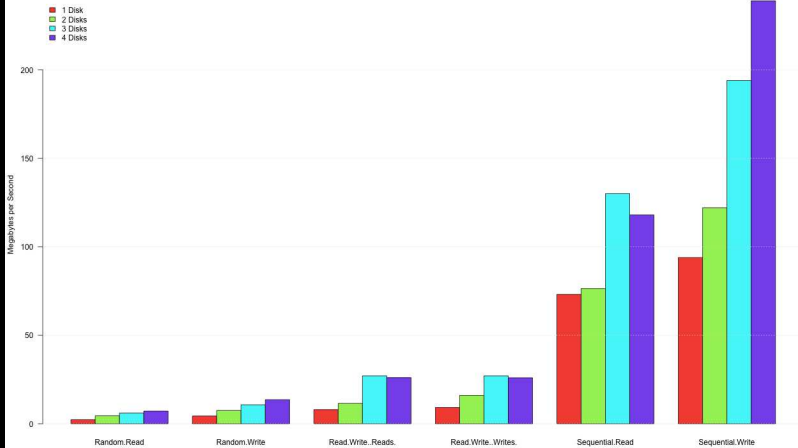
2.6.28-gentoo

ext4 RAID 0



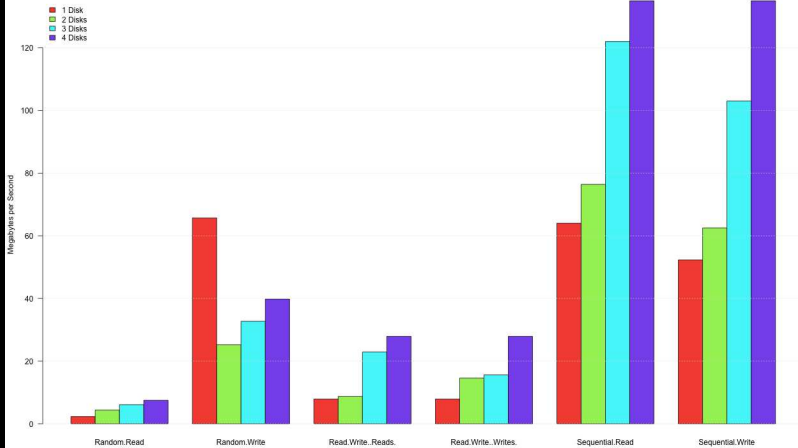
2.6.28-gentoo

jfs RAID 0



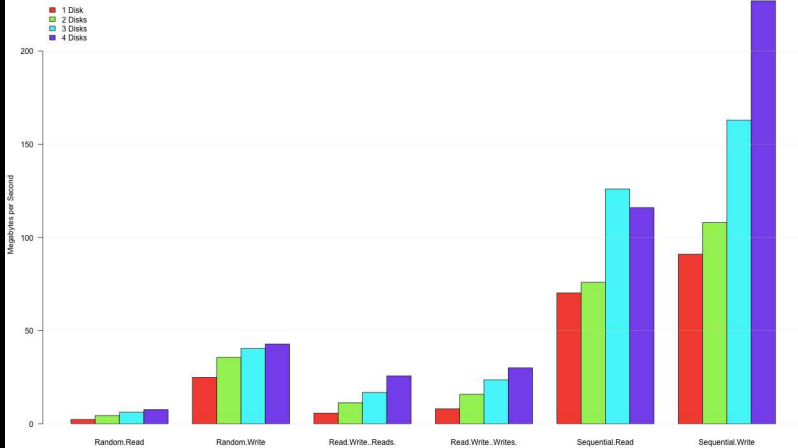
2.6.28-gentoo

reiserfs RAID 0



2.6.28-gentoo

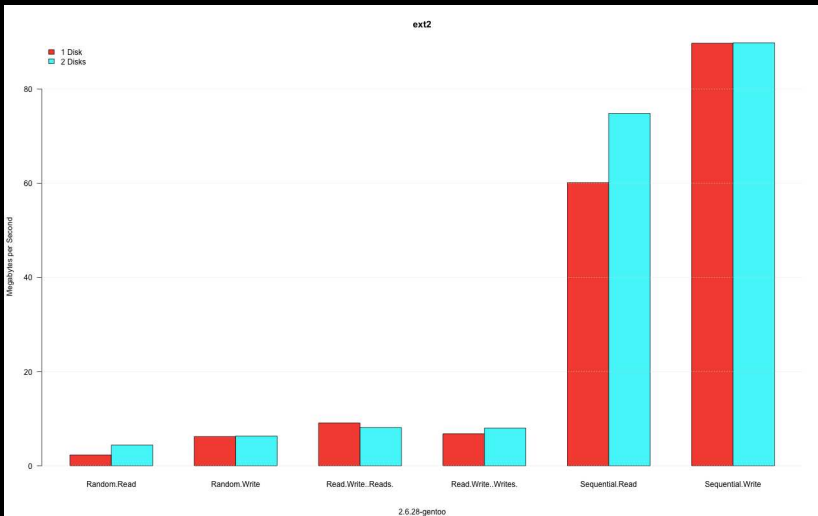
xfs RAID 0



2.6.28-gentoo

Note trends might not be the same when adding disks to other RAID configurations.

What benefits are there in mirroring?



- ▶ Should have same write performance as a single disk.
- ▶ May be able to take advantage of using two disks for reading.

What is the best use of 4 disks?

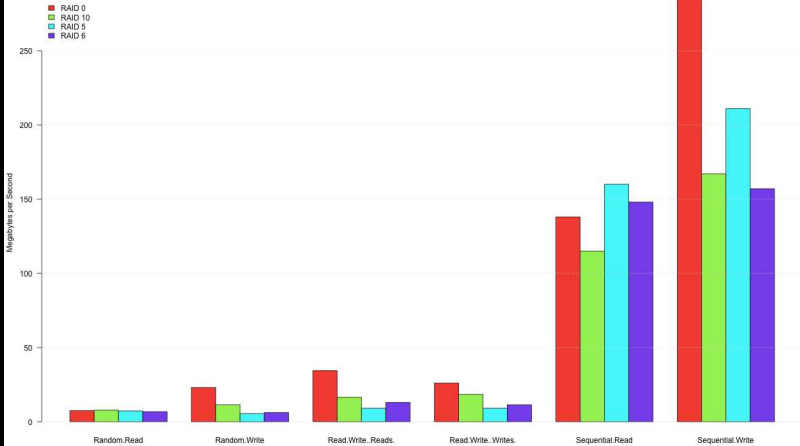
Is RAID-6 and improvement over RAID-5?²

- ▶ RAID 5 (striped disks with parity) combines three or more disks in a way that protects data against loss of any one disk; the storage capacity of the array is reduced by one disk.
- ▶ RAID 6 (striped disks with dual parity) (less common) can recover from the loss of two disks.

Do the answers depend on what filesystem is used?

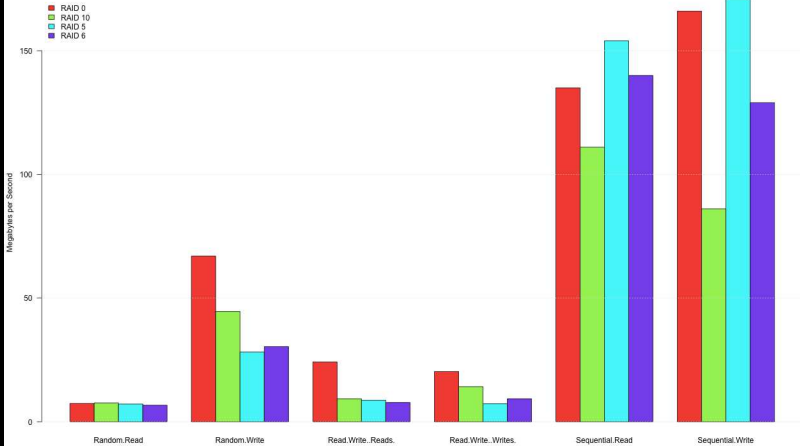
²<http://en.wikipedia.org/wiki/RAID>

ext2

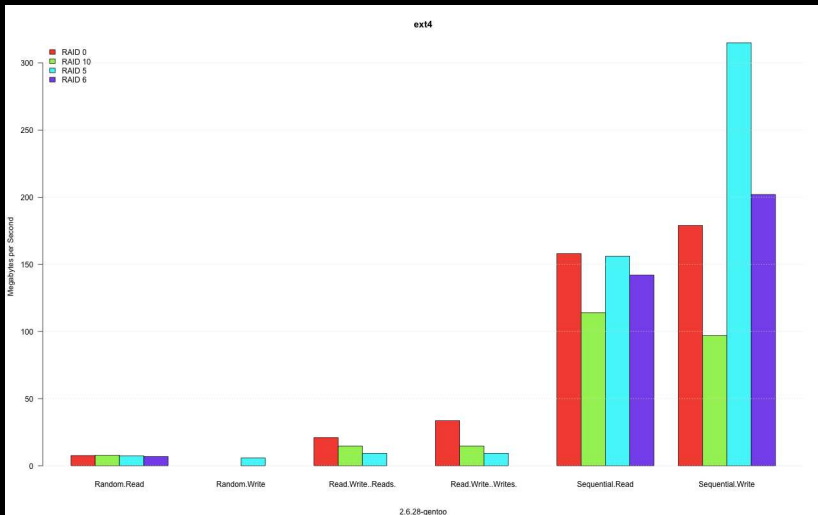


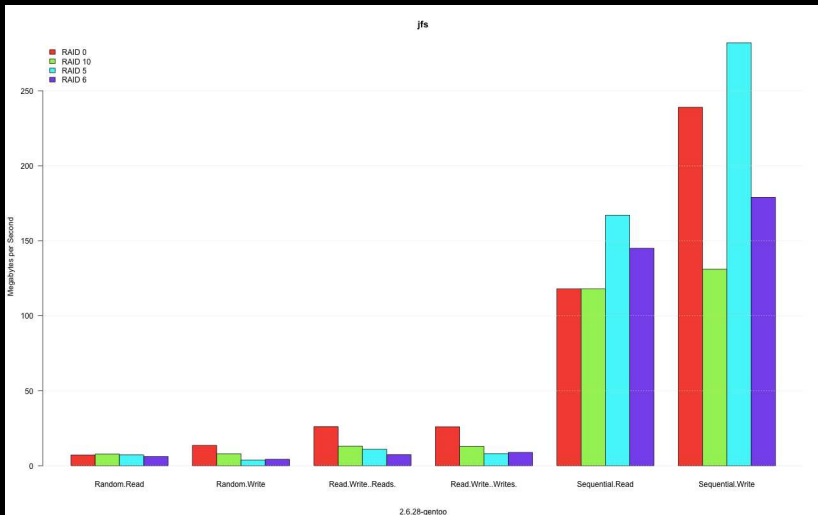
2.6.28-gentoo

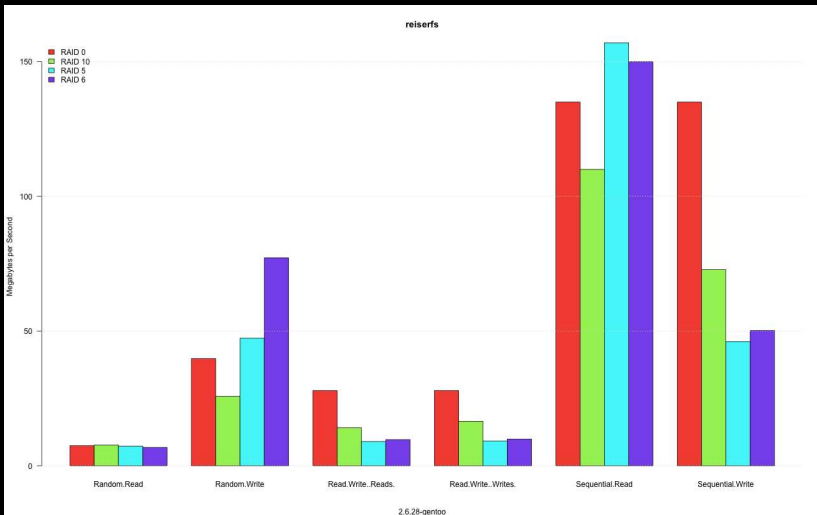
ext3

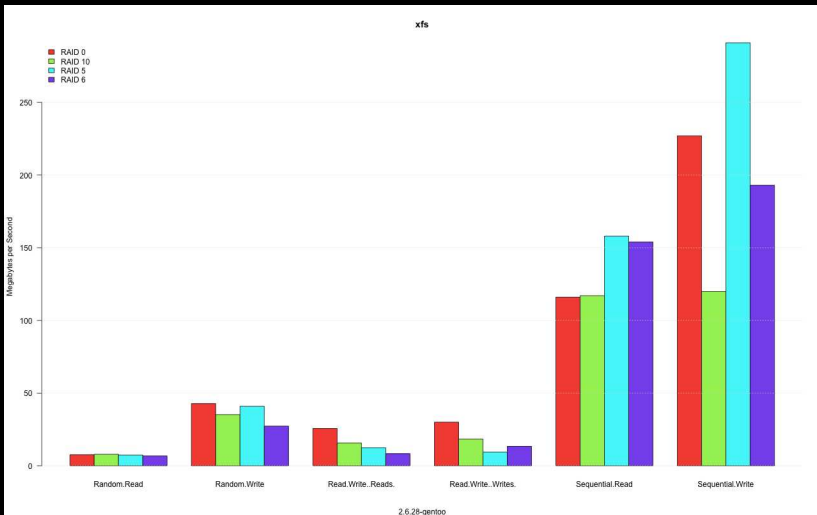


2.6.28-gentoo









Filesystem Readahead

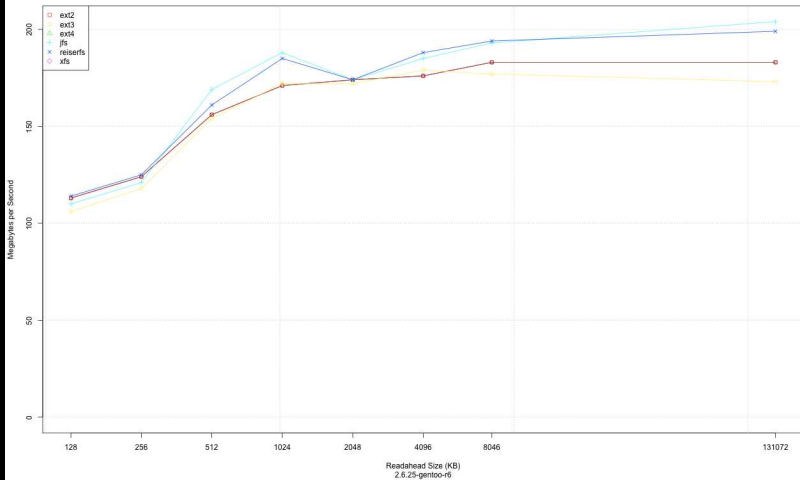
Values are in number of 512-byte segments, set per device.
To get the current readahead size:

```
$ blockdev --getra /dev/sda  
256
```

To set the readahead size to 8 MB:

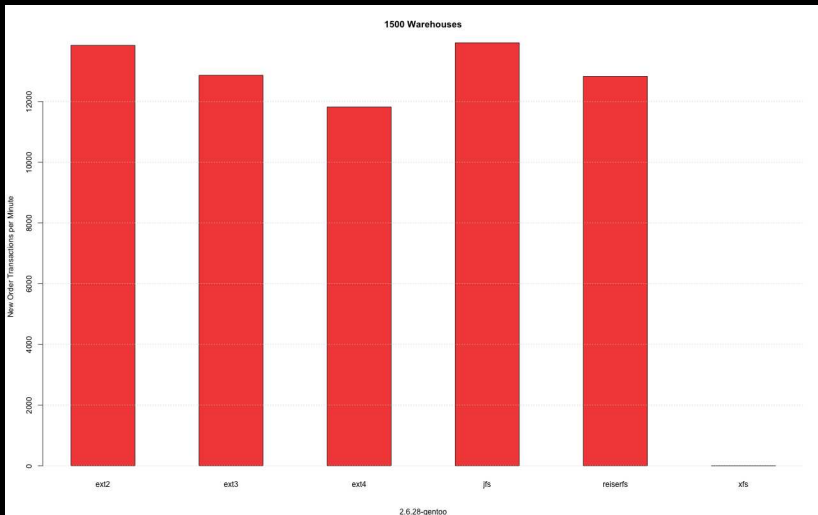
```
$ blockdev --setra 16384 /dev/sda
```

3 Disk RAID 5 Sequential Read Throughput



What does this all mean for my database system?

It depends... Let's look at one sample of just using difference filesystems.



Raw data for comparing filesystems

Initial inspection of results suggests we are close to being bound by processor and storage performance:

- ▶ <http://207.173.203.223/~markwkm/community6/dbt2/m-fs/ext2.1500.2/>
- ▶ <http://207.173.203.223/~markwkm/community6/dbt2/m-fs/ext3.1500.2/>
- ▶ <http://207.173.203.223/~markwkm/community6/dbt2/m-fs/ext4.1500.2/>
- ▶ <http://207.173.203.223/~markwkm/community6/dbt2/m-fs/jfs.1500.2/>
- ▶ <http://207.173.203.223/~markwkm/community6/dbt2/m-fs/reiserfs.1500.2/>
- ▶ <http://207.173.203.223/~markwkm/community6/dbt2/m-fs/xfs.1500.2/>

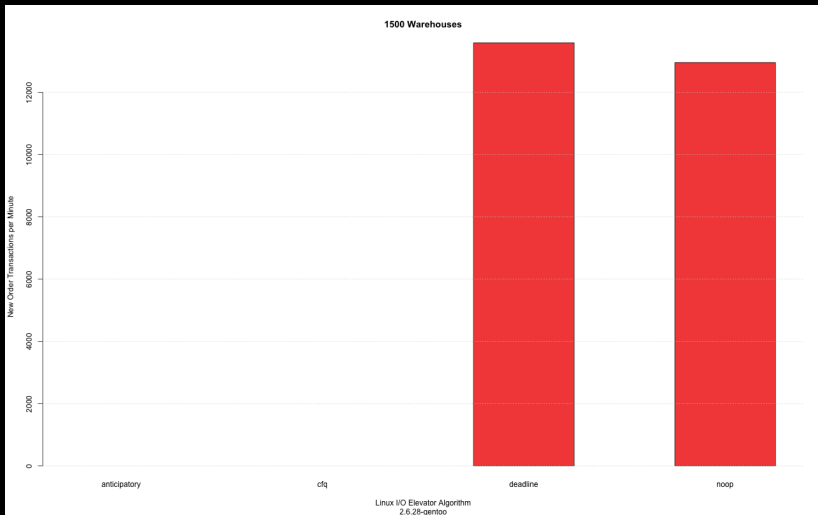
Linux I/O Elevator Algorithm

To get current algorithm in []'s:

```
$ cat /sys/block/sda/queue/scheduler  
noop anticipatory [deadline] cfq
```

To set a different algorithm:

```
$ echo cfg > /sys/block/sda/queue/scheduler  
noop anticipatory [deadline] cfq
```



Raw data for comparing I/O elevator algorithms

Initial inspection of results suggests we are close to being bound by processor and storage performance:

- ▶ <http://207.173.203.223/~markwkm/community6/dbt2/m-elevator/ext2.anticipatory.1/>
- ▶ <http://207.173.203.223/~markwkm/community6/dbt2/m-elevator/ext2.cfs.1/>
- ▶ <http://207.173.203.223/~markwkm/community6/dbt2/m-elevator/ext2.deadline.1/>
- ▶ <http://207.173.203.223/~markwkm/community6/dbt2/m-elevator/ext2.noop.1/>

Why use deadline?³

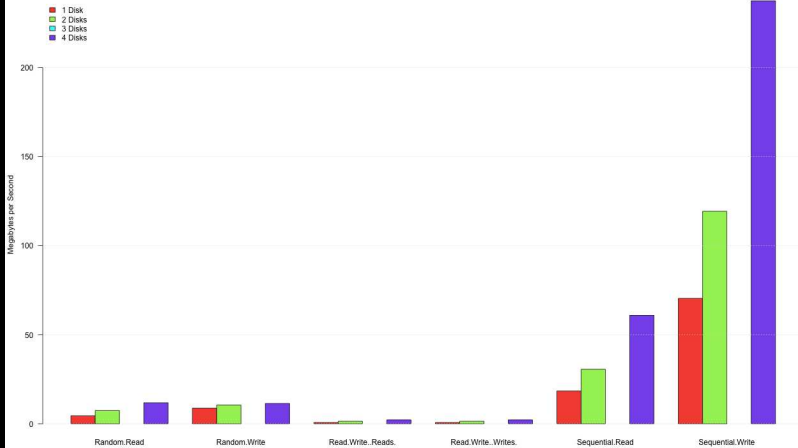
- ▶ Attention! Database servers, especially those using "TCQ" disks should investigate performance with the 'deadline' IO scheduler. Any system with high disk performance requirements should do so, in fact.
- ▶ Also, users with hardware RAID controllers, doing striping, may find highly variable performance results with using the as-iosched. The as-iosched anticipatory implementation is based on the notion that a disk device has only one physical seeking head. A striped RAID controller actually has a head for each physical device in the logical RAID device.

³Linux source code: [Documentation/block/as-iosched.txt](#)

Bonus Data - Preview of FreeBSD

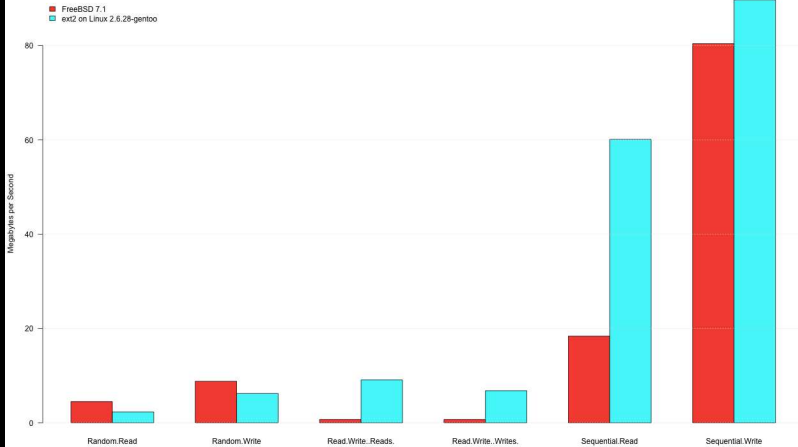
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```

ufs RAID 0



FreeBSD 7.1

1 Disk RAID 0



Final Notes

- ▶ You have to test your own system to see what it can do.

Materials Are Freely Available - In a new location!

PDF

- ▶ <http://www.slideshare.net/markwkm>

L^AT_EX Beamer (source)

- ▶ <http://git.postgresql.org/git/performance-tuning.git>

Time and Location

When: 2nd Thursday of the month

Location: Portland State University

Room: FAB 86-01 (Fourth Avenue Building)

Map: <http://www.pdx.edu/map.html>

Coming up next time... Effects of tuning PostgreSQL Parameters

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```

Acknowledgements

Haley Jane Wakenshaw

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```

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